Data Science in the AI Era

How should data science education evolve?

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Keynote, Universidad Externado de Colombia
October 21, 2026
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UC Berkeley UC Berkeley College of Computing, Data Science, and Society

Outline

1. Data science

What a decade of teaching data science at Berkeley has taught us.

2. AI

What changes when models can write code and analyze data.

3. Colombia

Open-source contributors and questions from Colombian data.

4. Evolving

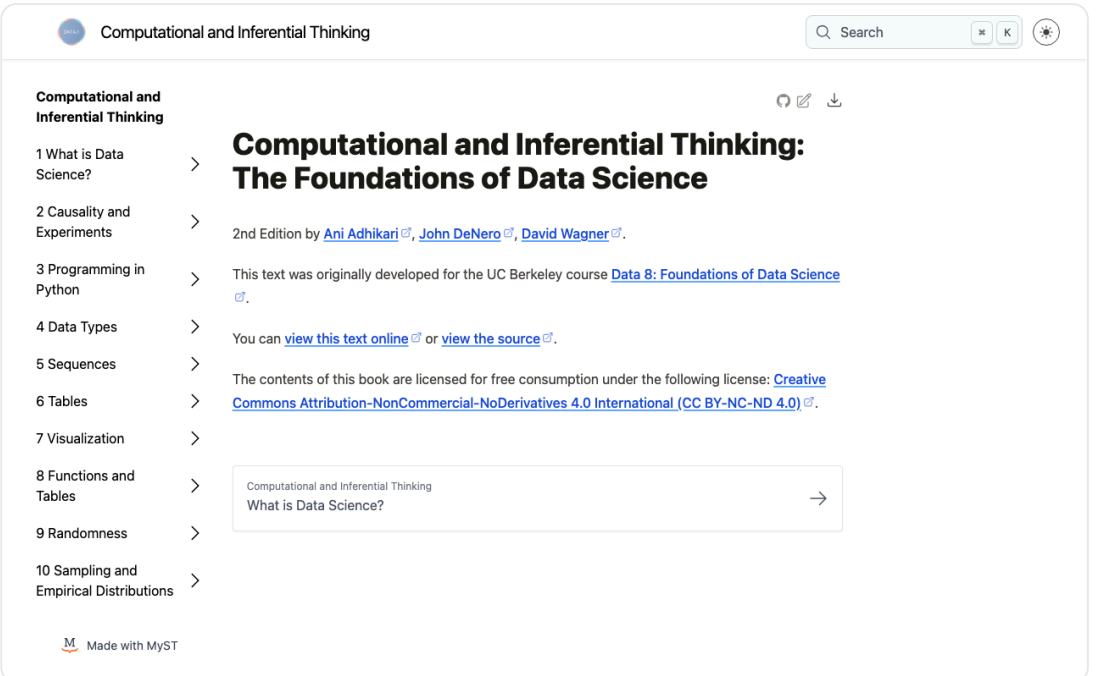
How courses and students can grow, and why there is more to do.

> If AI makes computational tools available to more people, what new questions can students investigate?

The Data 8 story

A first course in computing and statistics, for any major, in the browser, with real data from day one.

> The largest course at UC Berkeley. Up to half of undergraduates take it, and it is the gateway to Berkeley's largest major.



inferentialthinking.com

> Students did not have to learn all of the machinery before they could do data science.

The notebook became the laboratory

Browser → Notebook → Data → Experiment

- Jupyter, Python and real data
- Change the code and see what happens
- The same environment for instructors and students
- Reproducible across a class
- Works for 20 students or for thousands
- Open-source tools, open curriculum and shared infrastructure

> Next, I would like to see notebooks in the mathematics foundation as well: calculus, linear algebra and discrete mathematics.

Three lessons from a decade of scaling

Remove friction

JupyterHub means no installs and no laptop gap. Students meet ideas before they meet configuration problems.

Connect to disciplines

Connector courses such as Data 88E (Economic Models) give students the domain context that makes methods useful.

Open everything

Open textbook, notebooks, autograding and deployment on GitHub. Openness is what let the curriculum travel.

> These lessons are how Data 8 reached community colleges in California and partners such as Universidad Externado.

Interdisciplinarity in the Data Science major

Berkeley built disciplinary expertise into the Data Science major through the Domain Emphasis requirement.

Economics	Biology	Physics	Public policy

- Students connect computational skills with substantive knowledge in a field.
- AI is becoming relevant to research and practice across disciplines.

A new field, with a curriculum built fresh

Data Science did not inherit a curriculum. Berkeley had to design one, with faculty from statistics, computer science and many other disciplines.

Interdisciplinary faculty

Courses designed by teams from statistics, computer science and the domains.

Open materials

Textbooks, notebooks and assignments kept on GitHub, where they can be revised each term.

Continual renewal

Data 8 and Data 100 are moving from pandas to Polars. Notebooks are being updated for AI tools.

> A young field can keep renewing its curriculum as the tools change.

What if we did the same thing with AI?

Put a model in the notebook, and let students investigate it the way they investigate data.

- Prompt it
- Change parameters
- Examine tokens and probabilities
- Create embeddings
- Add retrieval
- Compare models
- Build evaluations
- Investigate failures

The notebook can become a laboratory for AI.

Who owns the laboratory?

Commercial AI

Notebook

↓

API

↓

Proprietary model

↓

Commercial cloud

Students learn to use a product.

Open-weight models on an open stack

Notebook

↓

Open-source tooling

↓

Open model

↓

Shared or public compute

Students can inspect, modify, evaluate and build.

Small open models as teaching tools

A small open model is a file that students can count, inspect and test. Many experiments do not need a frontier model.

- Parameters × bits ÷ 8 = size
- Tokens and next-token probabilities
- Temperature
- The linear algebra underneath

```python
from transformers import AutoTokenizer
from transformers import AutoModelForCausalLM

name = "HuggingFaceTB/SmolLM2-135M"  # 270 MB
tok = AutoTokenizer.from_pretrained(name)
lm = AutoModelForCausalLM.from_pretrained(name)

prompt = "Bogotá is famous for its"
ids = tok(prompt, return_tensors="pt")
probs = lm(**ids).logits[0, -1].softmax(-1)
for p, i in zip(*probs.topk(5)):
    print(tok.decode(i), f"{p:.0%}")
```



Small models on a JupyterHub

In [10]:
```
# set the model path
model_path = "shared-rw/qwen2-1_5b-instruct-q4_0.gguf"
# example model path, change to your model location
```

In [11]:
```
# Load the model from shared storage
model = Llama(
   model_path=model_path ,
    n_ctx=2048,      # Context window size (how much text the model can process at once)
    verbose=False    # Set to True if you want to see detailed loading information
)

print("Model loaded successfully!")
```

```
Model loaded successfully!
```

Generating Text

Now let's try generating some text. We'll give the model a prompt and see what it produces.

In [12]:
```
# Generate text from a simple prompt
import textwrap

prompt = "Python is a programming language that"

output = model(
    prompt,
    max_tokens=100,      # Maximum length of generated text
    temperature=0.7,     # Controls randomness (0 = deterministic, 1+ = creative)
    stop=["\n\n"]        # Stop generation at double newline
)

# Extract the generated text
generated_text = output['choices'][0]['text']
print(f"Prompt: {prompt}")
print(f"Generated:\n{textwrap.fill(generated_text, width=80)}")
```

```
Prompt: Python is a programming language that
Generated:  allows you to do many different things. It can be used for web development, data analys
is, machine learning, and more. It's also a great language for beginners because it's easy to learn
and has a lot of great resources online to help you get started. Python is also widely used in scie
ntific research, as it's a powerful tool for data analysis and machine learning.
```

The model is stored in a shared folder on the JupyterHub. Students load it, change parameters and observe the results, with no setup.

- Administrator: sets up the environment
- Instructor: sets up the curriculum
- Student: explores

Cloudbank Classroom

Small models in the browser



Try it on your laptop

ericvd-ucb.github.io/small-models-lite

jupyterlite

A JupyterLite site where small language models run in the student's browser through WebLLM, with no server, API key or installation.

- Talking to models and generation settings
- Tokens, chunking and embeddings
- Memory with SQLite and embedding search
- Inside an open-weight model package

Requires Chrome or Edge with WebGPU.

Why small open models?

Environment and carbon

Data centres used about 485 TWh last year, roughly as much as Germany, and the IEA expects that to double by 2030. Laptops and campus servers use little power and need little cooling.

Sovereignty

Models on institutional servers keep control of the tools, the data and privacy. Open-weight models trail the frontier by about four months.

Less corporate control over education

Courses built on open models do not depend on a company's pricing, terms of service or product decisions. Subsidized AI for rent cannot last indefinitely.

> "Educators can train students not only how to use AI models, but also how to determine when local models are preferable to cloud-based systems."

Buhler, Pérez & Boettiger, "Scientists should shift away from AI data centres," *Nature* 656, 296–298 (2026)

Fernando Pérez and IPython

- From Medellín; studied at Universidad de Antioquia
- PhD in Physics at CU Boulder, where he took time from his dissertation to write IPython
- Professor of Statistics at UC Berkeley
- Now leading work to bring Jupyter to new fields
  - GIS, biology and health



Fernando Pérez
fperez

Follow

Open source, open science, AI in science for earth, healthcare. **@ipython** creator, **@jupyter** and **@2i2c-org** co-founder. Prof at UC Berkeley, **@BIDS** & **@SchmidtDSE**.

github.com/fperez

Juanita Gómez: a path through open source

- Undergraduate in pure mathematics at Universidad Javeriana, Colombia
- Core developer for the Spyder IDE at Quansight
- Community manager for the Scientific Python Project
- PhD candidate in Computer Science at UC Santa Cruz
- Open Source Sustainability Program Manager at NVIDIA



Juanita Gomez
juanis2112

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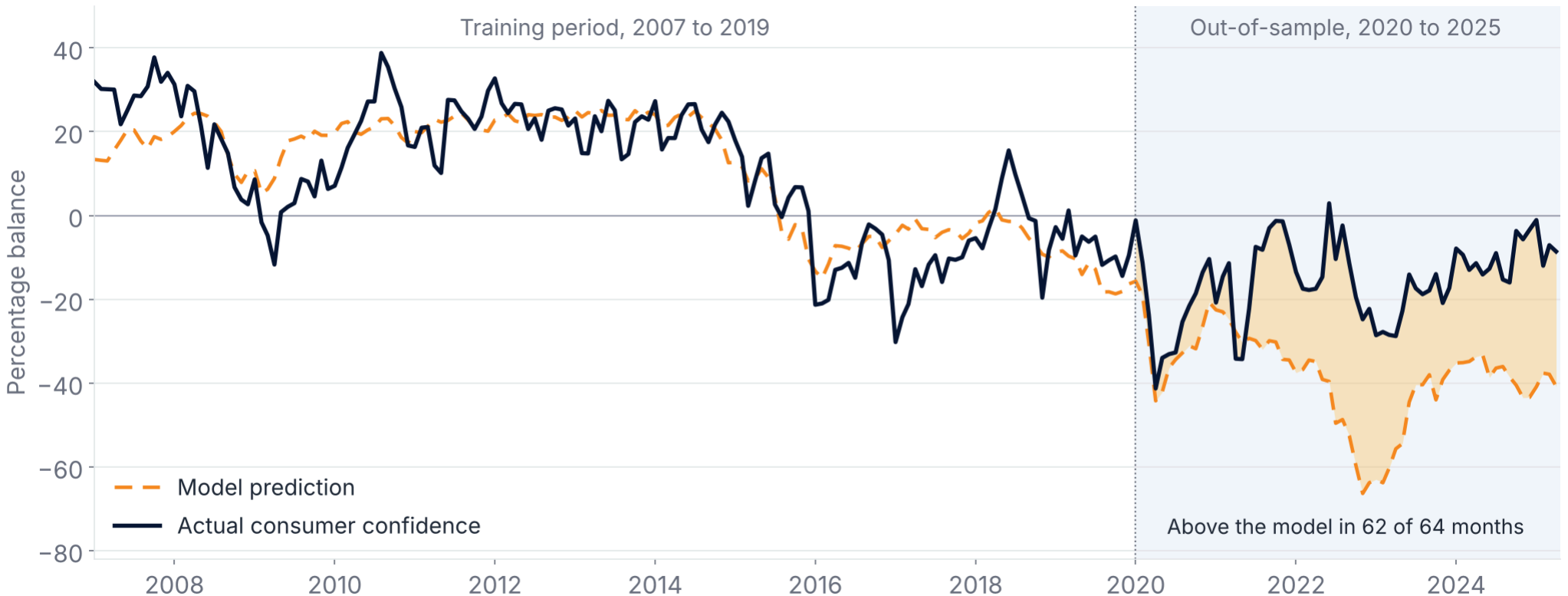
Everyone can participate in open source

Open-source projects depend on participants, and many contributions do not require writing code.

- **Code**
- **Documentation**
- **Bug reports**
- **Tutorials**
- **Answering questions**
- **Supporting new users**

> My own outreach to educators has made me a member of the Jupyter community. The open-source communities around small models have made learning about AI much easier.

A Colombian question: the vibecession



Consumer confidence (Fedesarrollo, via OECD and FRED) and an OLS prediction from inflation, unemployment, the interbank rate, pesos per dollar and the stock index, trained on 2007 to 2019.

> After 2020, Colombians were more optimistic than this model predicts. Without the exchange rate in the model, the gap reverses sign.

Shared tools, local questions

Eight notebooks prepared for students in Colombia, each running in Google Colab or JupyterLite:

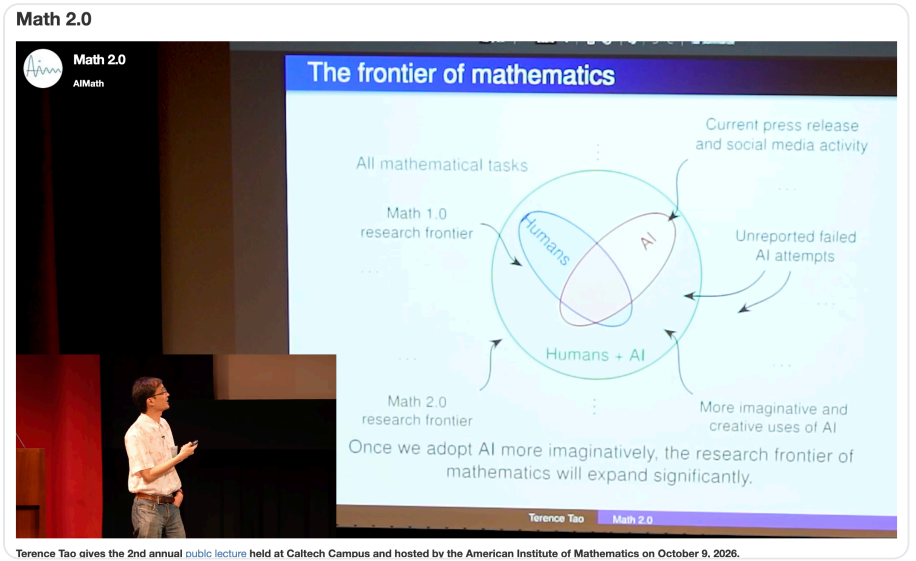
- US and Colombian vibecessions
- Colombia's economy by president
- Bollinger bands
- ML trading: ARIMA, XGBoost and Kronos
- Fantasy Premier League team selection
- SQL with Formula 1 data



github.com/ericvd-ucb/Colombia-26-NB

> Open materials allow instructors to adapt a curriculum to local data rather than start from scratch.

Terence Tao: the frontier keeps expanding



Terence Tao, "Math 2.0," AIM Public Lecture, Caltech, October 2026. Video

> "We are now entering an era of proof abundance in mathematics."
>
> "The future frontiers of mathematics will always continue to expand."

The same holds for data science:

- Instructors can build richer labs and explorations
- Students can take on more ambitious projects
- More questions become feasible, in more fields

AI does not leave less data science to do. It leaves more.

Expanding what students learn

SQL course

Add data engineering: pipelines, transformations and data quality.

Machine learning course

Add MLOps: pipelines, deployment, evaluation and monitoring.

Labs

Instructors can build richer explorations with real data.

Projects

Students can build larger projects in the same semester.

> AI tools reduce the time needed for some technical steps. That time can go to topics we could not previously fit into a course.

More projects, more active learning

As AI tools take over some routine steps, courses can shift time from passive learning to projects and interactive work.

Less time on

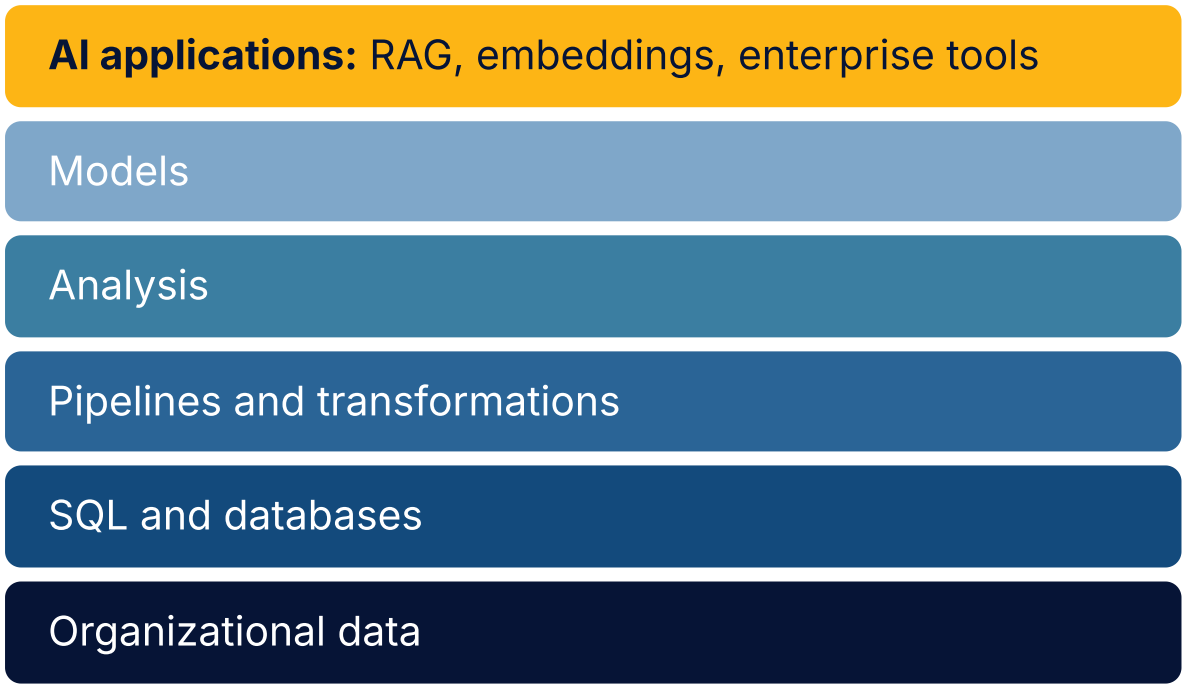
- Lectures students only watch
- Code copied from worked examples
- Problem sets with one right answer

More time on

- Projects with real data and open questions
- Notebooks students modify and extend
- Explaining, presenting and reviewing work

> How should we assess learning when students work alongside AI tools?

Data engineering as a foundation for ML and AI



Many Data Science graduates begin professional work with SQL, databases and data transformations.

AI may automate some query writing. Understanding an organization's data and keeping pipelines reliable remain essential.

From the Data Science lifecycle to MLOps

Data Science lifecycle	AI harness engineering
Collect data	Parse data into a tokenized format
Prepare	Data pipelines
Model	Inference pipelines
Evaluate	Parse and manage outputs
Iterate	Update and iterate

> Data Science curricula can build on this foundation by introducing MLOps and AI operations.

Students need to see the whole stack

A student can understand gradient descent but have no idea how an AI application works.

A student can now build an AI application without understanding gradient descent.

Both are educational problems.

- Data engineering
- APIs
- Containers and reproducible environments
- Cloud, GPUs and inference
- MLOps and LLMOps
- Evaluation and monitoring

> How much engineering should a Data Science curriculum teach, alongside mathematics and statistics?

Fundamentals and applications

There is a tension between learning fundamentals and learning applications. Data Science courses have shown that students can learn both at once.

Fundamentals

- Inference and uncertainty
- Probability and statistics
- Linear algebra
- Computational thinking

Real tools

- Jupyter and Python
- pandas, Polars and scikit-learn
- SQL and databases
- Git and GitHub

> In Data 8, the tools are introduced in service of the concepts, using real data from the first week.

Learning a new Python package

Read the docs → Supply data → Configure → Run → Interpret

- In Econ 148, students meet new data and ML packages every few weeks and learn to use them productively.
  - NumPy → statsmodels → scikit-learn
- AI tools expose similar Python interfaces and APIs, and the same pattern applies.
  - One Econ 148 lecture uses nine packages, from statsmodels ARIMA to the Kronos foundation model and an LLM API.











Debugging and evaluating AI workflows

Debugging is a large part of using AI tools. Problems can come from any part of a workflow, not only the code.

Data quality	Joins	Transformations
Code	Models	Interpretation

- Check assumptions, inspect intermediate results and question unexpected outputs.
- AI-generated code and model outputs still require validation.
- Test-driven development and reproducibility.

Domain expertise as a comparative advantage

- AI assistants are making coding and prototyping more accessible.
- Identifying worthwhile problems requires knowing a field, its institutions and its constraints.
- Domain expertise helps decide which data matter and whether results are credible.

Students who combine domain knowledge, Data Science and AI fluency can build distinctive applications.

Professional responsibility in AI

Data and AI systems operate within social and institutional contexts. Data Science graduates can help shape how they are evaluated and governed.

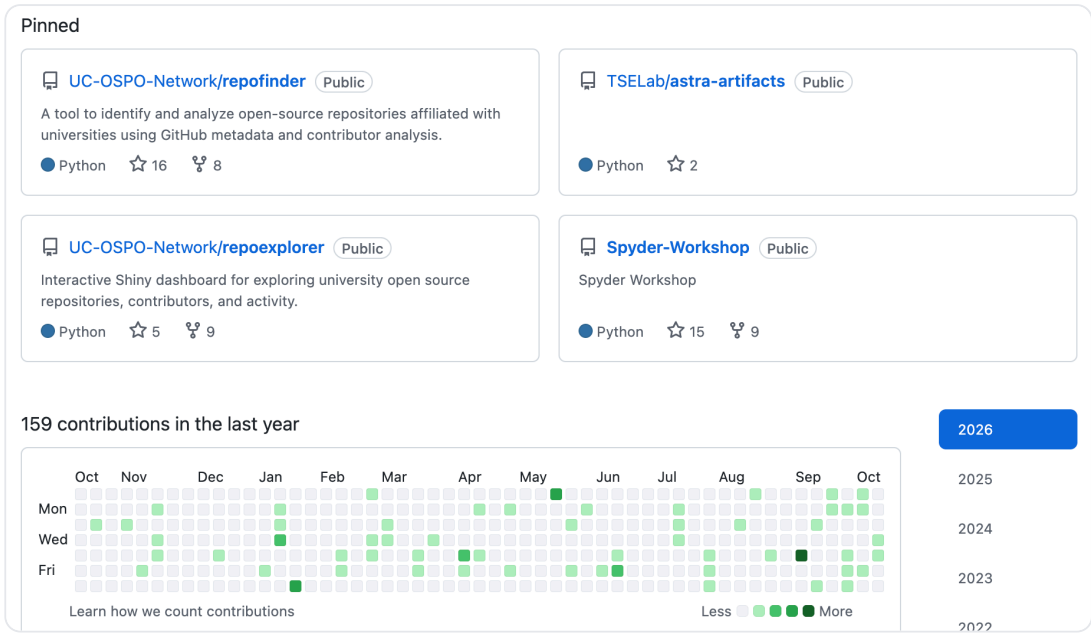
- Evaluate performance, reliability and limitations
- Test products for bias and unequal impacts
- Protect privacy; promote documentation, reproducibility and auditing
- Help build professional norms and safeguards
- Recognize when to raise concerns or challenge decisions

> What should a professional do when they understand a system well enough to see that it may cause harm?

Building a GitHub portfolio

A portfolio documents what you built and how you thought about it.

- Pinned projects, each with a short description
- A record of contributions over time
- Work in other people's projects and organizations



github.com/juanis2112

> A well-explained project tells an employer more than code alone.

Data science in the AI era

Learn to compute

Computational literacy belongs in every discipline.

Learn to question

Statistics, inference and domain knowledge decide whether a result is credible.

Learn to build

Contribute to open-source communities and work on problems that matter to you.

Ten years ago we asked how every student could learn with data. Now we can ask how every student can look inside AI, and how much more they can do with it.

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